

Project management

Chapter 5 of Ian Sommerville's
Software Engineering

Objectives

- To explain the main tasks undertaken by project managers
- To introduce software project management and to describe its distinctive characteristics
- To discuss project planning and the planning process
- To show how graphical schedule representations are used by project management
- To discuss the notion of risks and the risk management process

Topics covered

- Management activities
 - Project planning
 - Project scheduling
 - Risk management
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Software project management

- Concerned with activities involved in ensuring that software is delivered **on time** and **on schedule** and in accordance with the **requirements** of the organisations developing and procuring the software.
- Project management is needed because software development is always subject to **budget and schedule constraints** that are set by the organisation developing the software.

Software management distinctions

- The product is **intangible**.
 - The product is uniquely **flexible**.
 - Software engineering is **not recognized as an engineering discipline** with the same status as mechanical, electrical engineering, etc.
 - The software development process is **not standardised**.
 - Many software projects are **'one-off' projects**.
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Management activities

- Proposal writing.
 - Project planning and scheduling.
 - Project costing.
 - Project monitoring and reviews.
 - Personnel selection and evaluation.
 - Report writing and presentations.
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Management commonalities

- These activities are **not peculiar** to software management.
- Many techniques of engineering project management are **equally applicable** to software project management.
- Technically complex engineering systems tend to suffer from the **same problems** as software systems.

Project staffing

- May **not** be possible to appoint the **ideal people** to work on a project
 - Project **budget may not allow** for the use of highly-paid staff;
 - Staff with the **appropriate experience may not be available**;
 - An organisation may wish to **develop employee skills** on a software project.
- Managers have to work within these constraints especially when there are **shortages of trained staff**.

Project planning

- Probably the most **time-consuming** project management activity.
 - **Continuous activity** from initial concept through to system delivery. Plans must be regularly **revised** as new information becomes available.
 - Various **different types of plan** may be developed to support the main software project plan that is concerned with schedule and budget.
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Types of project plan

Plan		Description
Quality plan		Describes the quality procedures and standards that will be used in a project. See Chapter 27.
Validation plan		Describes the approach, resources and schedule used for system validation. See Chapter 22.
Configuration management plan		Describes the configuration management procedures and structures to be used. See Chapter 29.
Maintenance plan		Predicts the maintenance requirements of the system, maintenance costs and effort required. See Chapter 21.
Staff development plan.		Describes how the skills and experience of the project team members will be developed. See Chapter 25.

Project planning process

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Establish the project constraints
Make initial assessments of the project parameters
Define project milestones and deliverables
while project has not been completed or cancelled loop
    Draw up project schedule
    Initiate activities according to schedule
    Wait ( for a while )
    Review project progress
    Revise estimates of project parameters
    Update the project schedule
    Re-negotiate project constraints and deliverables
    if ( problems arise ) then
        Initiate technical review and possible revision
    end if
end loop
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The project plan

- The project plan sets out:
 - The **resources** available to the project;
 - The work **breakdown**;
 - A **schedule** for the work.
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Project plan structure

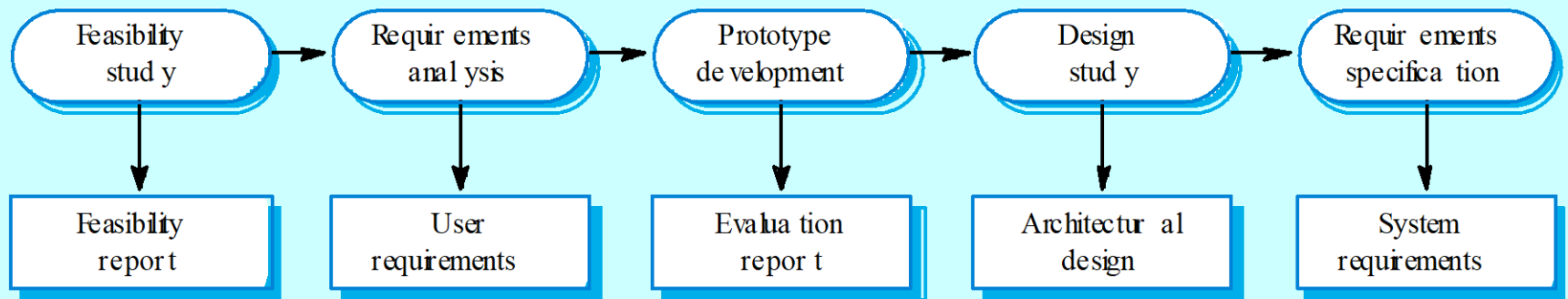
- Introduction.
 - Project organisation.
 - Risk analysis.
 - Hardware and software resource requirements.
 - Work breakdown.
 - Project schedule.
 - Monitoring and reporting mechanisms.
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Activity organization

- Activities in a project should be organised to produce **tangible outputs** for management to **judge progress**.
 - **Milestones** are the end-point of a process activity.
 - **Deliverables** are project results delivered to customers.
 - The **waterfall** process allows for the straightforward definition of progress milestones.
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Milestones in the RE process

ACTIVITIES

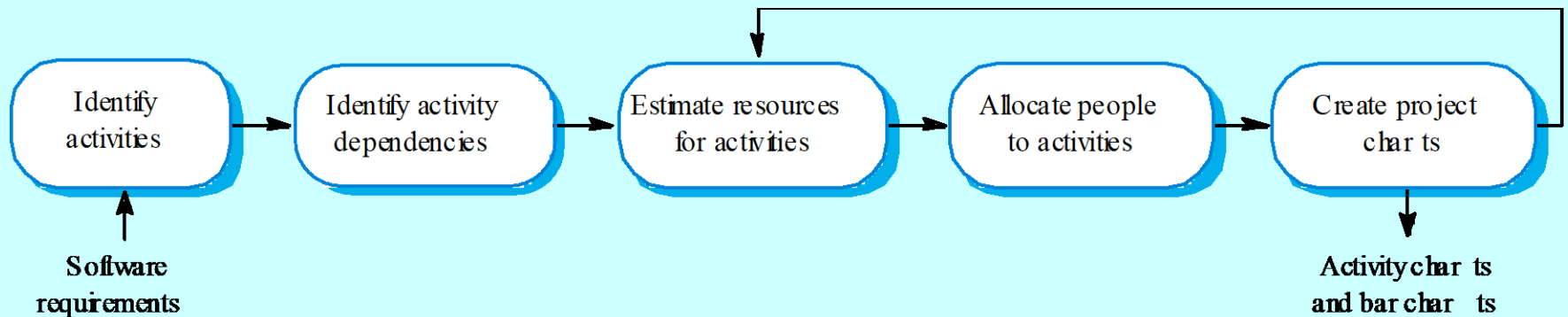


MILESTONES

Project scheduling

- Split project into **tasks** and **estimate time** and **resources** required to complete each task.
- Organize tasks **concurrently** to make optimal use of workforce.
- **Minimize task dependencies** to avoid delays caused by one task waiting for another to complete.
- Dependent on **project managers intuition** and **experience**.

The project scheduling process



Scheduling problems

- **Estimating the difficulty** of problems and hence the cost of developing a solution is hard.
- **Productivity is not proportional** to the number of **people** working on a task.
- Adding people to a late project makes it later because of **communication overheads**.
- The unexpected always happens. Always allow **contingency in planning**.

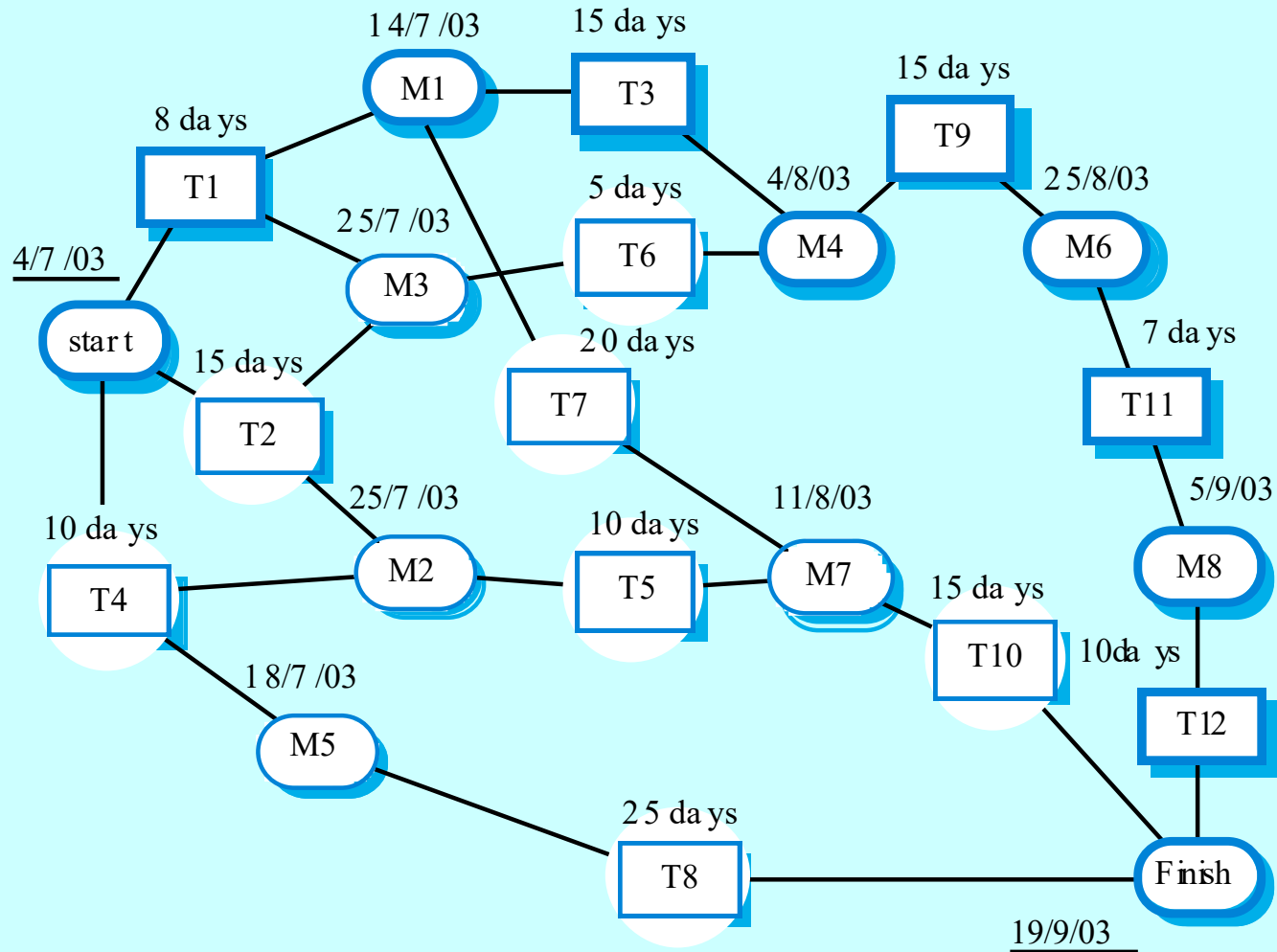
Bar charts and activity networks

- Graphical notations used to illustrate the project schedule.
- Show project breakdown into tasks. Tasks should not be too small. They should take about a week or two.
- Activity charts show task dependencies and the critical path.
- Bar charts show schedule against calendar time.

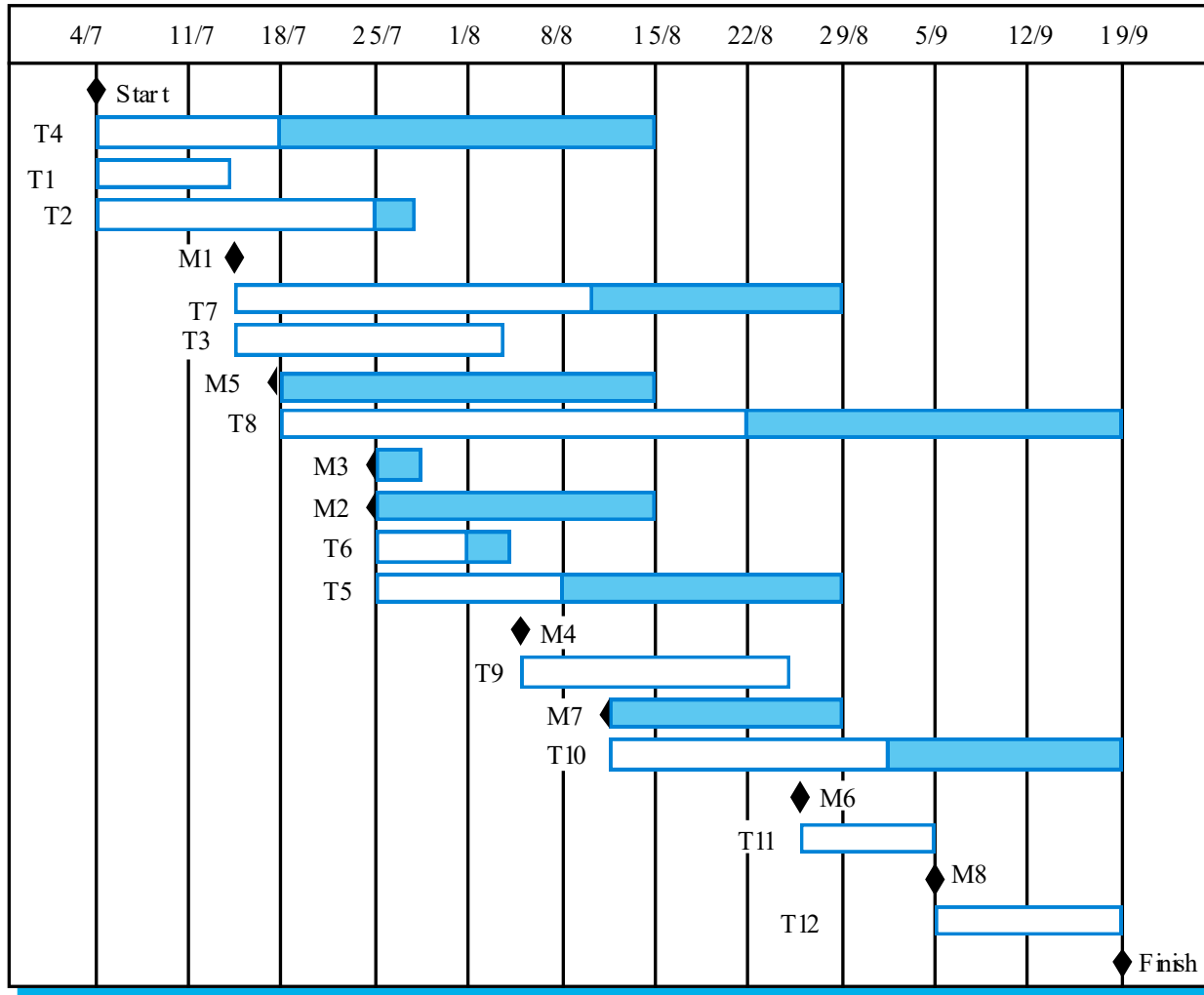
Task durations and dependencies

Activity	Duration (days)	Dependencies
T1	8	
T2	15	
T3	15	T1 (M1)
T4	10	
T5	10	T2, T4 (M2)
T6	5	T1, T2 (M3)
T7	20	T1 (M1)
T8	25	T4 (M5)
T9	15	T3, T6 (M4)
T10	15	T5, T7 (M7)
T11	7	T9 (M6)
T12	10	T11 (M8)

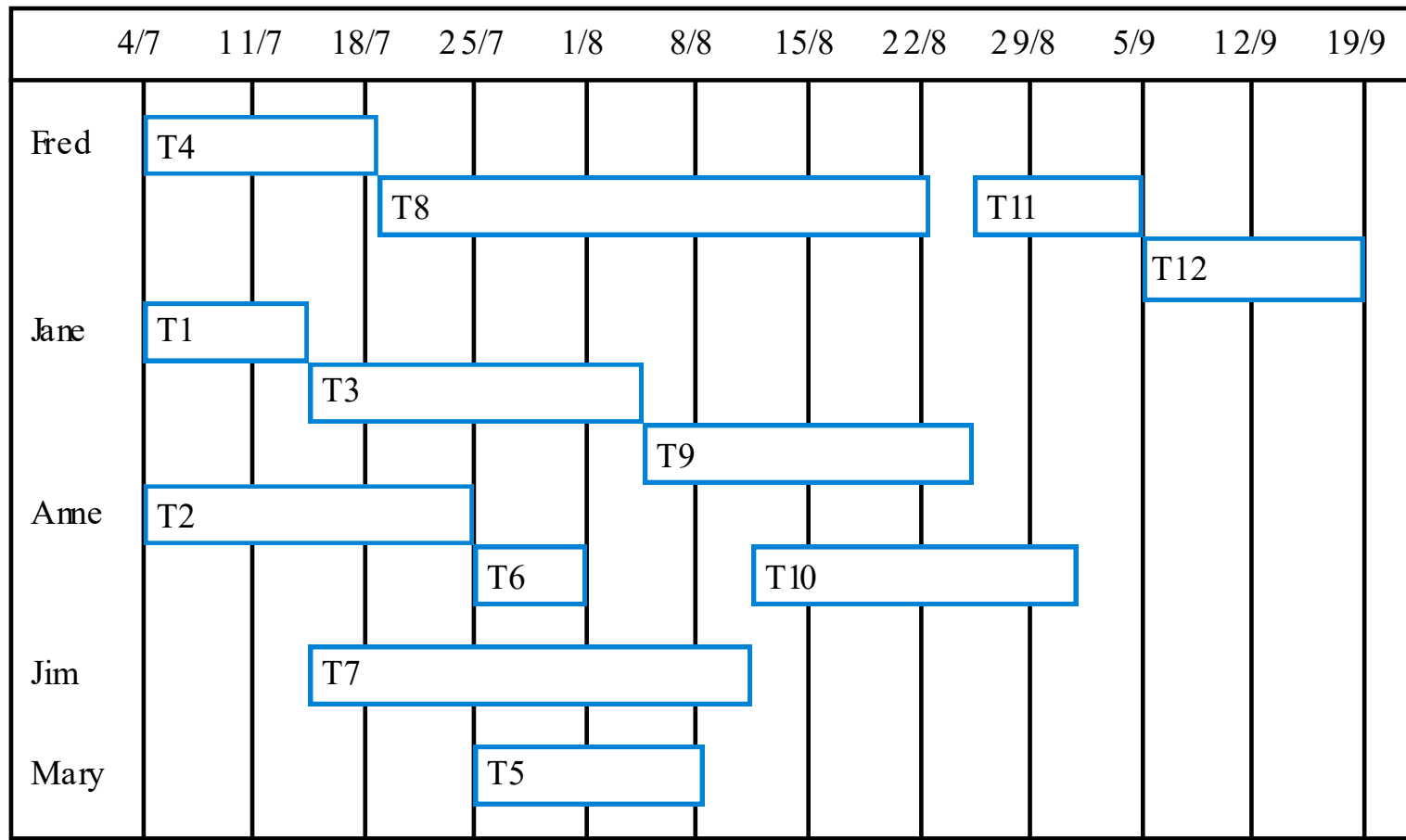
Activity network



Activity timeline



Staff allocation



Risk management

- Risk management is concerned with identifying risks and drawing up plans to minimise their effect on a project.
 - A risk is a probability that some adverse circumstance will occur
 - Project risks affect schedule or resources;
 - Product risks affect the quality or performance of the software being developed;
 - Business risks affect the organisation developing or procuring the software.
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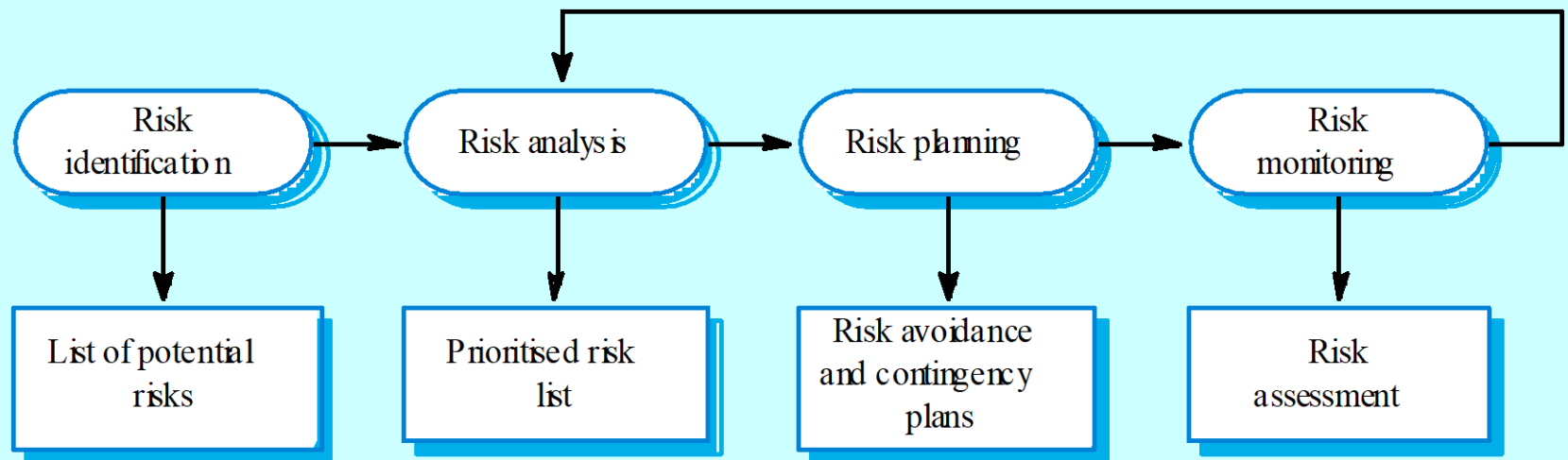
Software risks

Risk	Affects	Description
Staff turnover	Project	Experienced staff will leave the project before it is finished.
Management change	Project	There will be a change of organisational management with different priorities.
Hardware unavailability	Project	Hardware that is essential for the project will not be delivered on schedule.
Requirements change	Project and product	There will be a larger number of changes to the requirements than anticipated.
Specification delays	Project and product	Specifications of essential interfaces are not available on schedule
Size underestimate	Project and product	The size of the system has been underestimated.
CASE tool under-performance	Product	CASE tools which support the project do not perform as anticipated
Technology change	Business	The underlying technology on which the system is built is superseded by new technology.
Product competition	Business	A competitive product is marketed before the system is completed.

The risk management process

- Risk **identification**
 - Identify project, product and business risks;
 - Risk **analysis**
 - Assess the likelihood and consequences of these risks;
 - Risk **planning**
 - Draw up plans to avoid or minimise the effects of the risk;
 - Risk **monitoring**
 - Monitor the risks throughout the project;
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The risk management process



Risk identification

- Technology risks.
 - People risks.
 - Organisational risks.
 - Requirements risks.
 - Estimation risks.
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Risks and risk types

Risk type	Possible risks
Technology	The database used in the system cannot process as many transactions per second as expected. Software components that should be reused contain defects that limit their functionality.
People	It is impossible to recruit staff with the skills required. Key staff are ill and unavailable at critical times. Required training for staff is not available.
Organisational	The organisation is restructured so that different management are responsible for the project. Organisational financial problems force reductions in the project budget.
Tools	The code generated by CASE tools is inefficient. CASE tools cannot be integrated.
Requirements	Changes to requirements that require major design rework are proposed. Customers fail to understand the impact of requirements changes.
Estimation	The time required to develop the software is underestimated. The rate of defect repair is underestimated. The size of the software is underestimated.

Risk analysis

- Assess probability and seriousness of each risk.
 - Probability may be very low, low, moderate, high or very high.
 - Risk effects might be catastrophic, serious, tolerable or insignificant.
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Risk analysis (i)

Risk	Probability	Effects
Organisational financial problems force reductions in the project budget.	Low	Catastrophic
It is impossible to recruit staff with the skills required for the project.	High	Catastrophic
Key staff are ill at critical times in the project.	Moderate	Serious
Software components that should be reused contain defects which limit their functionality.	Moderate	Serious
Changes to requirements that require major design rework are proposed.	Moderate	Serious
The organisation is restructured so that different management are responsible for the project.	High	Serious

Risk analysis (ii)

Risk	Probability	Effects
The database used in the system cannot process as many transactions per second as expected.	Moderate	Serious
The time required to develop the software is underestimated.	High	Serious
CASE tools cannot be integrated.	High	Tolerable
Customers fail to understand the impact of requirements changes.	Moderate	Tolerable
Required training for staff is not available.	Moderate	Tolerable
The rate of defect repair is underestimated.	Moderate	Tolerable
The size of the software is underestimated.	High	Tolerable
The code generated by CASE tools is inefficient.	Moderate	Insignificant

Risk planning

- Consider each risk and develop a strategy to manage that risk.
- **Avoidance strategies**
 - The probability that the risk will arise is reduced;
- **Minimisation strategies**
 - The impact of the risk on the project or product will be reduced;
- **Contingency plans**
 - If the risk arises, contingency plans are plans to deal with that risk;

Risk management strategies (i)

Risk	Strategy
Organisational financial problems	Prepare a briefing document for senior management showing how the project is making a very important contribution to the goals of the business.
Recruitment problems	Alert customer of potential difficulties and the possibility of delays, investigate buying-in components.
Staff illness	Reorganise team so that there is more overlap of work and people therefore understand each other's jobs.
Defective components	Replace potentially defective components with bought-in components of known reliability.

Risk management strategies (ii)

Risk	Strategy
Requirements changes	Derive traceability information to assess requirements change impact, maximise information hiding in the design.
Organisational restructuring	Prepare a briefing document for senior management showing how the project is making a very important contribution to the goals of the business.
Database performance	Investigate the possibility of buying a higher-performance database.
Underestimated development time	Investigate buying in components, investigate use of a program generator

Risk monitoring

- Assess each identified risks regularly to decide whether or not it is becoming less or more probable.
 - Also assess whether the effects of the risk have changed.
 - Each key risk should be discussed at management progress meetings.
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Risk indicators

Risk type	Potential indicators
Technology	Late delivery of hardware or support software, many reported technology problems
People	Poor staff morale, poor relationships amongst team member, job availability
Organisational	Organisational gossip, lack of action by senior management
Tools	Reluctance by team members to use tools, complaints about CASE tools, demands for higher-powered workstations
Requirements	Many requirements change requests, customer complaints
Estimation	Failure to meet agreed schedule, failure to clear reported defects

System reliability specification

■ Hardware reliability

- What is the probability of a hardware component failing and how long does it take to repair that component?

■ Software reliability

- How likely is it that a software component will produce an incorrect output. Software failures are different from hardware failures in that software does not wear out. It can continue in operation even after an incorrect result has been produced.

■ Operator reliability

- How likely is it that the operator of a system will make an error?
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Reliability metrics

Metric	Explanation
POFOD Probability of failure on demand	The likelihood that the system will fail when a service request is made. A POFOD of 0.001 means that 1 out of a thousand service requests may result in failure.
ROCOF Rate of failure occurrence	The frequency of occurrence with which unexpected behaviour is likely to occur. A R OCOF of 2/100 means that 2 failures are likely to occur in each 100 operational time units. This metric is sometimes called the failure intensity.
MTTF Mean time to failure	The average time between observed system failures. An MTTF of 500 means that 1 failure can be expected every 500 time units.
AVAIL Availability	The probability that the system is available for use at a given time. Availability of 0.998 means that in every 1000 time units, the system is likely to be available for 998 of these.

Probability of failure on demand

- This is the probability that the system will fail when a service request is made. Useful when demands for service are intermittent and relatively infrequent.
 - Appropriate for protection systems where services are demanded occasionally and where there are serious consequence if the service is not delivered.
 - Relevant for many safety-critical systems with exception management components
 - Emergency shutdown system in a chemical plant.
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Rate of fault occurrence (ROCOF)

- Reflects the rate of occurrence of failure in the system.
 - ROCOF of 0.002 means 2 failures are likely in each 1000 operational time units e.g. 2 failures per 1000 hours of operation.
 - Relevant for operating systems, transaction processing systems where the system has to process a large number of similar requests that are relatively frequent
 - Credit card processing system, airline booking system.
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Mean time to failure

- Measure of the time between observed failures of the system. Is the reciprocal of ROCOF for stable systems.
 - MTTF of 500 means that the mean time between failures is 500 time units.
 - Relevant for systems with long transactions i.e. where system processing takes a long time. MTTF should be longer than transaction length
 - Computer-aided design systems where a designer will work on a design for several hours, word processor systems.
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Availability

- Measure of the fraction of the time that the system is available for use.
- Takes repair and restart time into account
- Availability of 0.998 means software is available for 998 out of 1000 time units.
- Relevant for non-stop, continuously running systems
 - telephone switching systems, railway signalling systems.

Non-functional requirements spec.

- Reliability measurements do NOT take the consequences of failure into account.
- Transient faults may have no real consequences but other faults may cause data loss or corruption and loss of system service.
- May be necessary to identify different failure classes and use different metrics for each of these. The reliability specification must be structured.

Failure consequences

- When specifying reliability, it is not just the number of system failures that matter but the consequences of these failures.
 - Failures that have serious consequences are clearly more damaging than those where repair and recovery is straightforward.
 - In some cases, therefore, different reliability specifications for different types of failure may be defined.
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Failure classification

Failure class	Description
Transient	Occurs only with certain inputs
Permanent	Occurs with all inputs
Recoverable	System can recover without operator intervention
Unrecoverable	Operator intervention needed to recover from failure
Non-corrupting	Failure does not corrupt system state or data
Corrupting	Failure corrupts system state or data

Steps to a reliability specification

- For each sub-system, analyse the consequences of possible system failures.
- From the system failure analysis, partition failures into appropriate classes.
- For each failure class identified, set out the reliability using an appropriate metric. Different metrics may be used for different reliability requirements.
- Identify functional reliability requirements to reduce the chances of critical failures.

Bank auto-teller system

- Each machine in a network is used 300 times a day
 - Bank has 1000 machines
 - Lifetime of software release is 2 years
 - Each machine handles about 200, 000 transactions
 - About 300, 000 database transactions in total per day
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Reliability specification for an ATM

Failure class	Example	Reliability metric
Permanent, non-corrupting.	The system fails to operate with any card that is input. Software must be restarted to correct failure.	ROCOF 1 occurrence/1000 days
Transient, non-corrupting	The magnetic stripe data cannot be read on an undamaged card that is input.	ROCOF 1 in 1000 transactions
Transient, corrupting	A pattern of transactions across the network causes database corruption.	Unquantifiable! Should never happen in the lifetime of the system

Specification validation

- It is impossible to empirically validate very high reliability specifications.
- No database corruptions means POFOD of less than 1 in 200 million.
- If a transaction takes 1 second, then simulating one day's transactions takes 3.5 days.
- It would take longer than the system's lifetime to test it for reliability.

Key points

- Good project management is **essential** for project success.
- The **intangible** nature of software causes problems for management.
- Managers have diverse roles but their most significant activities are **planning**, **estimating** and **scheduling**.
- Planning and estimating are **iterative** processes which continue throughout the course of a project.

Key points

- A **project milestone** is a predictable state where a formal report of progress is presented to management.
- **Project scheduling** involves preparing various graphical representations showing project activities, their durations and staffing.
- **Risk management** is concerned with identifying risks which may affect the project and planning to ensure that these risks do not develop into major threats.